Google Summer of Code 2026 Midterm Progress Report

Project

AI for Mental Health: Bias-Aware Crisis Signal Detection and Confidence Based Scoring System

Organization: HumanAI

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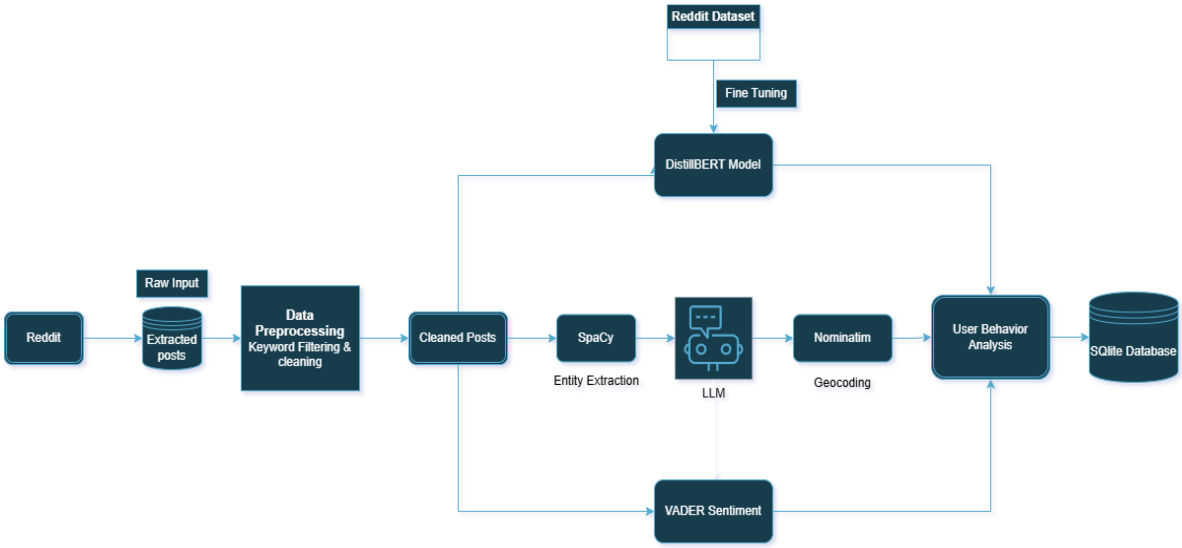
Evaluation: Midterm

1. Project Overview

During the first phase of the project, the primary objective was to establish a complete data processing pipeline. The implemented system automatically extracts Reddit posts, preprocesses textual content, classifies posts into Low-Risk, Moderate-Risk, and High-Risk categories using a fine-tuned DistilBERT model, performs sentiment analysis, extracts geographical references from post content, and generates user-level behavioral metrics. The processed information is stored Sqlite Database in a structured format to support downstream visualization and spatial analysis.

The outputs generated by this pipeline provide the foundation for the next stages of the project, which will focus on interactive dashboards, geospatial hotspot detection, confidence-based scoring, and bias-aware behavioral health monitoring. The completed backend establishes a scalable framework for transforming large-scale social media data into actionable insights for mental health research and crisis monitoring.

2. System Architecture



The proposed system is designed as a modular backend pipeline that transforms unstructured Reddit posts into structured behavioral health data for downstream spatial and behavioral analysis. The architecture consists of six major stages

- data collection
- data processing
- risk level and sentiment analysis
- geolocation extraction
- behavioral signal extraction
- structured data storage

3. Technologies Used

Category	Technologies Used
Programming Language	Python
Reddit Data Collection	PRAW (Reddit API)
Deep Learning Framework	PyTorch
Risk Classification	Fine-tuned DistilBERT (Hugging Face Transformers)
Sentiment Analysis	VADER Sentiment
Named Entity Recognition	spaCy
Contextual Location Inference	LangChain OpenAI
Geocoding	Nominatim, GeoPy
Data Processing	Pandas, NumPy
Database	SQLite
Data Visualization	Matplotlib, WordCloud, Folium

4. Implementation Details

4.1. Reddit Data Collection

The data collection pipeline utilizes the Reddit API through the PRAW library to retrieve publicly available posts from mental health-related subreddits. Along with the post content, relevant metadata including author information, subreddit, timestamp, score, and number of comments are collected. The extracted posts are stored as the raw input for the processing pipeline.

4.2. Data Preprocessing

The collected Reddit posts undergo a preprocessing stage to improve data quality before analysis. This stage includes text normalization, removal of URLs and unnecessary characters, whitespace normalization, duplicate removal, and keyword filtering along with coded language. The resulting cleaned text is used as the input for all downstream NLP modules.

4.3. Risk Level Classification

The risk classification module is responsible for identifying the severity of mental health concerns expressed in Reddit posts. A DistilBERT-based transformer model was fine-tuned to classify posts into three behavioral health risk categories: Low Risk, Moderate Risk, and High Risk.

Dataset Construction

To develop a robust multi-class classifier, a custom mental health risk dataset was created by combining and preprocessing two publicly available Kaggle datasets:

- **Sentiment Analysis for Mental Health**
- **Suicidal Tweet Detection Dataset**

The datasets were merged through the following preprocessing pipeline:

- Removal of missing and invalid records
- Text normalization and standardization
- Unified label schema
- Mapping of the original labels into three behavioral health risk categories

The final dataset consists of **47,894** annotated text samples distributed across three classes:

Risk Level	Description	Samples
High Risk	Explicit suicidal ideation, suicide planning, self-harm intent, or immediate crisis	12,064
Moderate Risk	Depression, anxiety, emotional distress, stress, or substance abuse without explicit suicidal intent	19,487
Low Risk	General discussions, daily life experiences, emotionally neutral, or non-crisis mental health conversations	16,343

Model Configuration

The classifier was developed by fine-tuning the **distilbert-base-uncased** model using the Hugging Face Transformers framework.

Epoch	Training Loss	Validation Loss	Accuracy	Weighted F1
1	0.2657	0.4282	85.45%	85.32%
2	0.2367	0.5043	85.80%	**85.71%**
3	0.1542	0.6890	85.79%	85.69%

The best-performing model achieved a validation accuracy of 85.80% and a weighted F1-score of 85.71%, demonstrating its effectiveness in distinguishing between varying levels of mental health risk.

4.4. Sentiment Analysis

To complement the risk classifier, VADER Sentiment Analysis is applied to every Reddit post. The sentiment analyzer generates polarity scores that are categorized as positive, negative, or neutral. These sentiment scores provide an additional behavioral signal and are later incorporated into the user behavior analysis module.

4.5. Geolocation Extraction

Accurately identifying user locations from Reddit posts presents a significant challenge because users frequently mention locations indirectly or through contextual descriptions rather than explicit place names. Applying a Large Language Model (LLM) to every Reddit post would improve contextual understanding but would also result in substantial computational and API costs.

To address this limitation, a **two-stage geolocation extraction strategy** is implemented.

In the first stage, **spaCy Named Entity Recognition (NER)** is used as a lightweight filtering mechanism to identify posts containing potential geographic entities. Posts without any location-related entities are excluded from further geolocation processing.

Only the filtered subset of posts is forwarded to the **OpenAI LLM**, which performs contextual reasoning to determine whether the extracted location refers to the user's own location rather than unrelated locations mentioned within the text. The LLM is also capable of inferring locations from contextual information that may not be directly represented as named entities.

The inferred locations are then geocoded using the **Nominatim** geocoding service to obtain standardized geographic coordinates (latitude and longitude). This hybrid approach significantly reduces the number of LLM API calls while maintaining high-quality location inference, making the system both scalable and cost-effective.

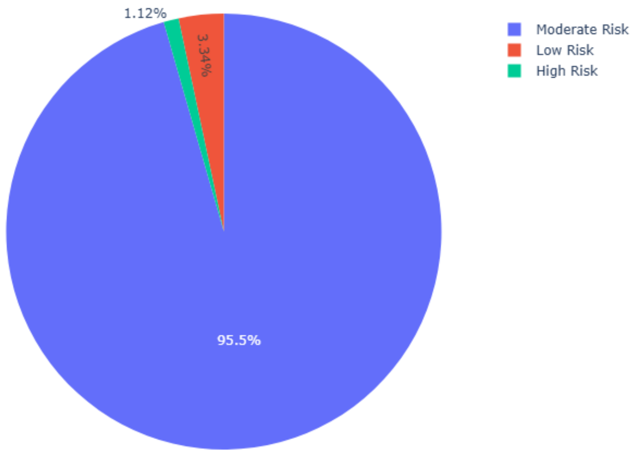
4.6. User Behaviour Analysis

The outputs from the risk classification, sentiment analysis, and geolocation extraction modules are integrated to generate user-level behavioral indicators. Additional features such as engagement statistics (upvotes and comments), posting frequency, temporal activity, and risk distribution are computed to provide a comprehensive representation of user behavior.

These aggregated behavioral features form the structured output of the backend pipeline and are stored in a SQLite database for subsequent visualization, geospatial analysis, and behavioral health monitoring.

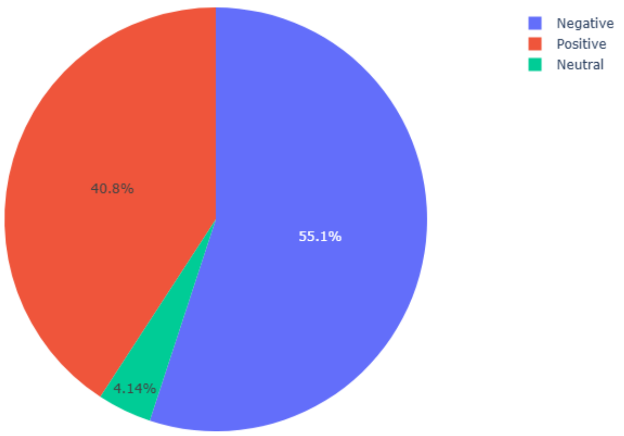
5. Results & Visualizations

5.1. Risk Level Distribution



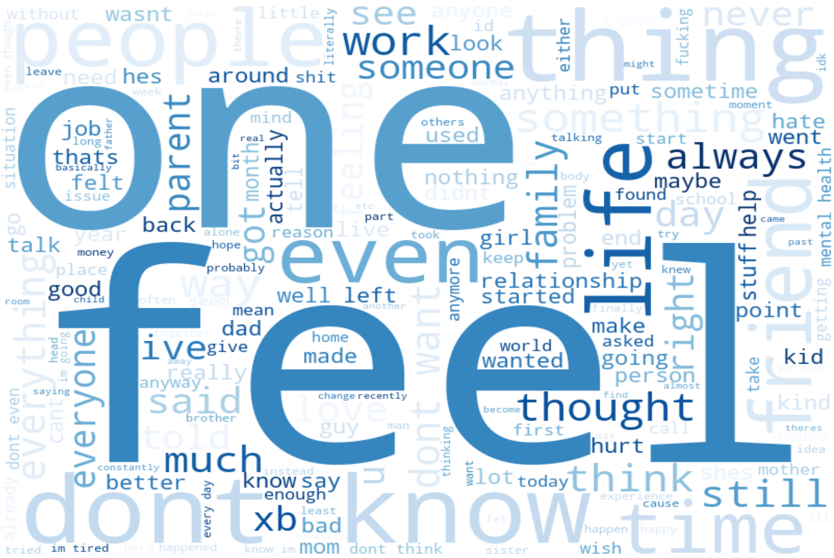
The majority of collected Reddit posts were classified as Moderate Risk (95.5%), followed by Low Risk (3.3%) and High Risk (1.1%).

5.2. Sentiment Distribution



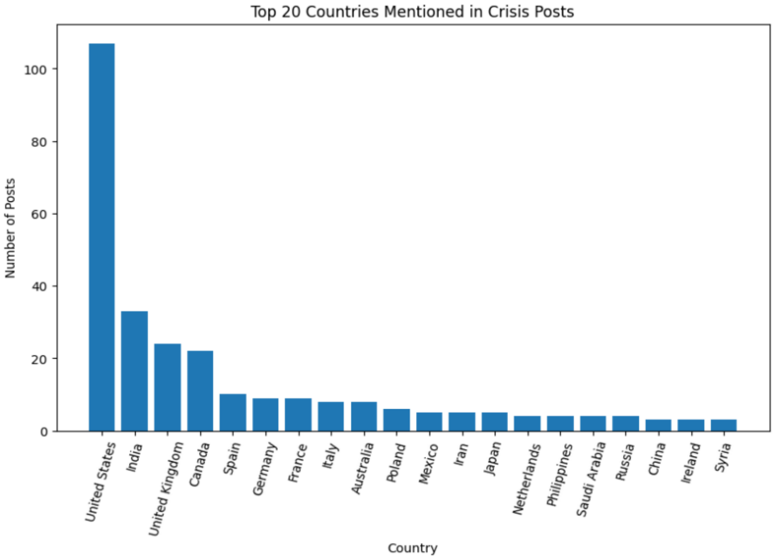
Negative sentiment dominates the collected posts, which aligns with the mental health-focused subreddits used during data collection. Sentiment serves as an additional behavioral signal alongside the risk classifier.

5.3. Word Cloud

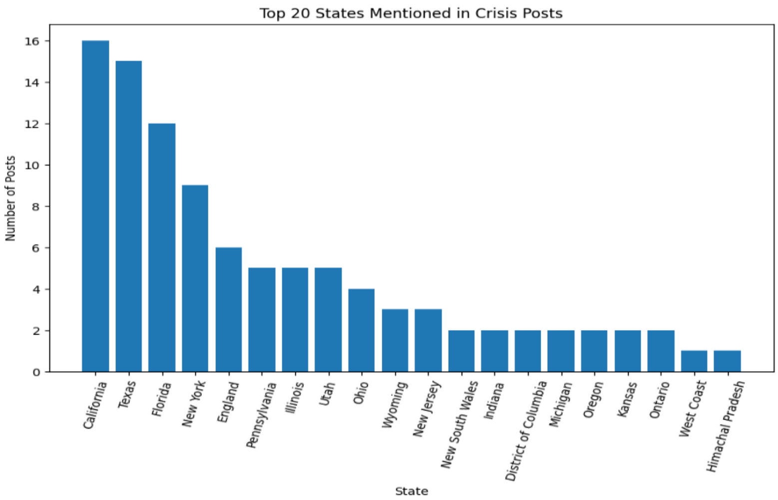


5.4. Geographical Distribution

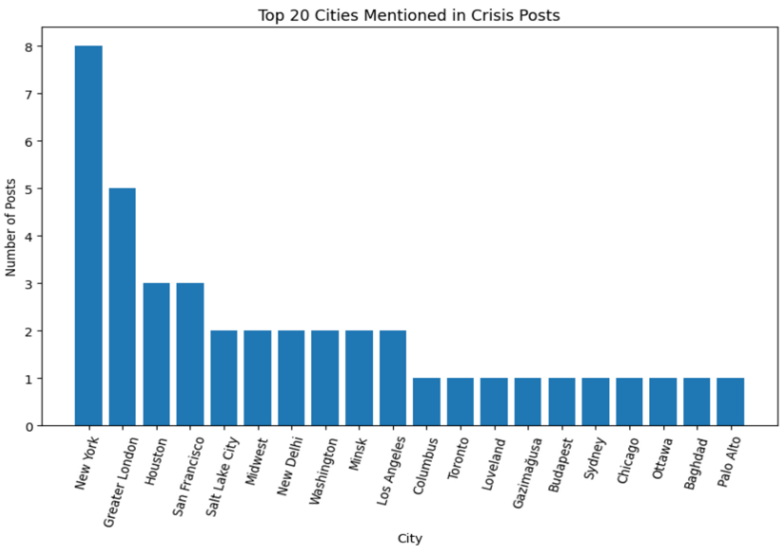
5.4.1. Top Countries Mentioned



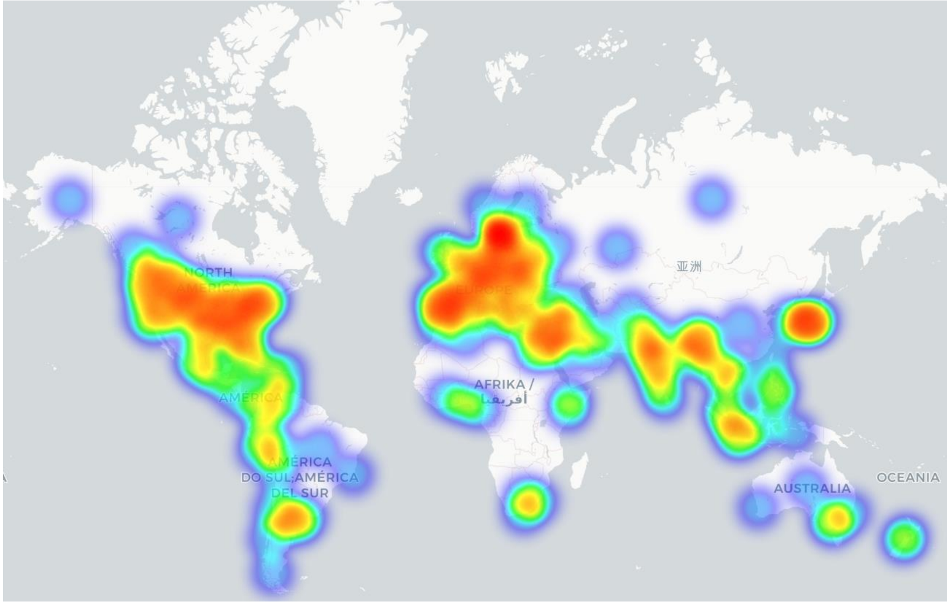
5.4.2. Top States Mentioned



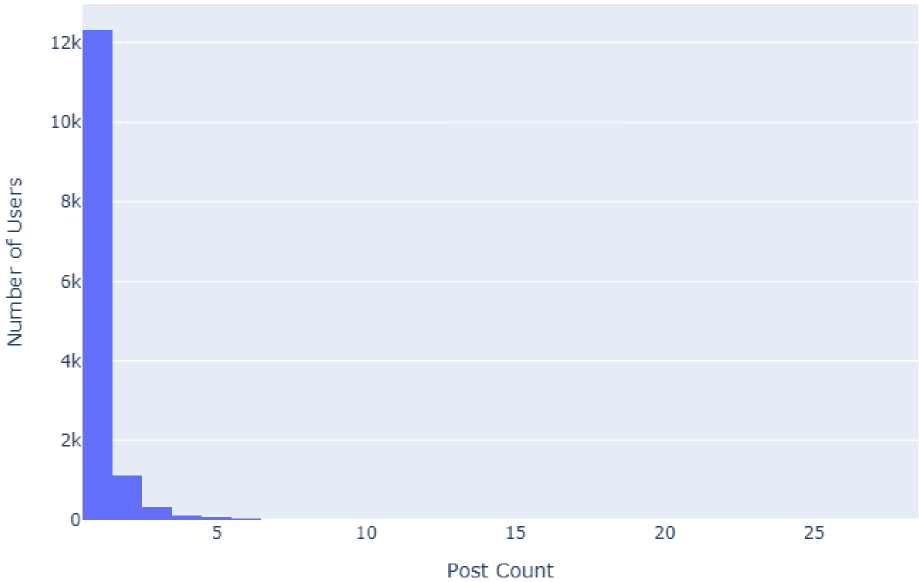
5.4.3. Top Cities Mentioned



5.5. Global Heatmap



5.6. User Activity Distribution



The majority of Reddit users contribute only one or two posts, while a small number of users exhibit substantially higher activity.

6. Challenges Encountered

6.1. Location Extraction Challenge

The primary challenge during the implementation was accurately extracting user location information from Reddit posts. An initial approach using spaCy Named Entity Recognition (NER) and Nominatim could identify explicit location mentions but struggled with indirect references, multiple locations in a single post, and determining whether a mentioned location actually referred to the user.

Applying a Large Language Model (LLM) to every post would improve contextual understanding but was not feasible due to API cost constraints. To address this, a hybrid approach was implemented where spaCy first filters posts containing potential location entities, and only those posts are processed by the OpenAI LLM for contextual location inference. The inferred locations are then geocoded using Nominatim. This approach improves location extraction accuracy while significantly reducing computational and API costs.

6.2. Dataset Construction for Risk Classification

Another challenge was the lack of a publicly available dataset that directly matched the project's three-level mental health risk classification (High Risk, Moderate Risk, and Low Risk). Existing datasets used different labeling schemes and were collected from different social media platforms. To address this, two publicly available datasets were combined, cleaned, relabeled, and standardized to create a unified three-class dataset suitable for training the DistilBERT classifier. This process involved duplicate removal, label mapping, text normalization, and quality filtering to ensure consistency across the merged dataset.

7. Future Work

The next phase of the project will focus on extending the completed backend pipeline into a comprehensive behavioral health monitoring platform. Planned work includes:

- Develop an interactive dashboard for visualizing risk levels, sentiment trends, user behavior, and geospatial distributions.
- Implement geospatial hotspot detection techniques to identify areas with elevated mental health risk.
- Develop a confidence-based scoring framework to quantify the reliability of model predictions and extracted location information.
- Incorporate bias-aware analysis to evaluate and mitigate potential demographic and geographic biases in the system.
- Optimize the geolocation extraction pipeline for improved accuracy and scalability.
- Conduct comprehensive evaluation and performance testing of the complete end-to-end system.
- Deploy the application for demonstration and support future research in AI-assisted mental health monitoring.

8. Conclusion

During the first phase of the Google Summer of Code project, a complete backend pipeline for behavioral health monitoring from Reddit data was successfully developed. The implemented system automates the workflow from Reddit post collection through preprocessing, risk level classification, sentiment analysis, geolocation extraction, and user behavior analysis, producing a structured dataset for downstream analytics.

In addition to developing the processing pipeline, a custom three-class mental health risk dataset was constructed and used to fine-tune a DistilBERT-based classifier, achieving a validation accuracy of 85.80% and a weighted F1-score of 85.71%. A cost-efficient hybrid geolocation extraction approach combining spaCy, an OpenAI LLM, and Nominatim was also implemented to improve location inference while minimizing API usage.

The completed backend will continue to the remaining phases of the project, including interactive dashboard, spatial hotspot detection, confidence-based scoring, and bias-aware behavioral health analysis.